Monocular Image-Based 3D Seated Posture Estimation Integrated with a Seat-Angle Adjustable Smart Chair

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Abstract: This paper proposes a monocular image-based 3D seated posture estimation method, integrates it into a smart chair with adjustable seat angle, and evaluates the system. First, an SMPL-based 3D body model is estimated from side RGB images using HMR2, and 3D skeleton keypoints excluding limb joints (SMPL NoLimbs) are extracted. Subsequently, a graph convolutional network classifies the seated posture into three classes (good, slightly poor, poor). Experiments using a large-scale dataset comprising 30 participants demonstrated that the proposed SMPL NoLimbs configuration outperformed silhouette images, 2D keypoints (YOLO), and 3D keypoints (VideoPose3D), achieving a macro F1 score of 0.595 for the three-class classification. Furthermore, we integrated the proposed method into a smart chair with a seat surface tiltable in multiple directions and angles, constructing and evaluating an integrated recognition-intervention seated support system. An evaluation experiment with 10 subjects demonstrated that the proposed system improved seated posture for many users and contributed to reducing pain and fatigue.

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1 INTRODUCTION

Prolonged sitting time has increased with the widespread adoption of desk work and remote meetings, leading to musculoskeletal disorders such as stiff shoulders and lower back pain. Epidemiological studies have demonstrated the negative health impact of prolonged sitting (Stamatakis et al., 2019), highlighting the importance of maintaining appropriate seated posture. Accordingly, smart chair systems aimed at promoting posture improvement and reducing physical burden have attracted increasing attention.

Many systems have been proposed to recognize seated posture using various sensing modalities, including cameras and pressure sensors, and to provide feedback or physical intervention for posture improvement (Odesola et al., 2024; Aminosharieh Najafi et al., 2022; Zhang et al., 2023). At the core of such systems lies seated posture recognition, for which accurate and real-time methods are essential.

Seated posture recognition approaches can be broadly categorized into vision-based and sensor-based methods. Vision-based approaches using monocular RGB images offer non-contact and low-cost solutions (Mallare et al., 2017; Kapoor et al., 2022; Li et al., 2023), whereas sensor-based approaches using pressure sensors often require hardware integration and maintenance, and may affect user comfort (Odesola et al., 2024). Moreover, many existing studies focus on classifying clearly defined or extreme postures, while the discrimination of subtle, natural seated postures observed during desk work remains insufficiently explored. In this study, we therefore target natural seated postures that users unconsciously adopt during desk work (Fig. 2) using monocular RGB images.

Most vision-based methods rely on 2D skeletal keypoints, and even 3D skeletons estimated from 2D keypoints (e.g., VideoPose3D) are limited to sparse joints that can be manually annotated (Kapoor et al., 2022; Li et al., 2023; Pavllo et al., 2019). As a result, detailed trunk-related features such as pelvic tilt and spinal curvature, which are critical for seated posture classification, are not sufficiently captured.

To address this limitation, we adopt an approach in which a parametric 3D human body model is first fitted to a monocular image, and dense skeletal keypoints are extracted from the reconstructed body model. Recent advances enable the estimation of SMPL parameters directly from images (Loper et al.,

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2015; Goel et al., 2023; Kolotouros et al., 2019; Saleem et al., 2025), allowing stable extraction of trunk-related 3D skeletal features including the spine and pelvis. Compared with conventional 2D or sparse 3D skeletons, SMPL-based keypoints provide a richer representation of trunk alignment, which is particularly suitable for discriminating subtle differences in seated posture.

Based on this idea, we use HMR2 (Goel et al., 2023) to estimate SMPL parameters from monocular images and extract 3D skeletal keypoints, which are then classified into three posture classes (Good, Slightly Poor, Poor) using a Graph Convolutional Network (GCN) (Kipf and Welling, 2016). The proposed method is trained and evaluated using a large-scale seated posture dataset collected from 30 participants.

In addition, few previous studies have jointly designed and evaluated the entire pipeline from posture recognition to feedback and physical intervention for posture improvement (Odesola et al., 2024), and recognition and intervention are often treated separately. Therefore, we integrate the proposed method into a seat-angle adjustable smart chair developed by Esumi et al. (Esumi et al., 2024), which can tilt the seat surface in anterior–posterior and left–right directions to modulate pelvic tilt and trunk alignment. We employ this smart chair as a physical intervention platform for recognition-driven posture guidance, as shown in Fig. 1, and demonstrate the feasibility of an integrated seated posture support system combining recognition and physical intervention.

The contributions of this study are summarized as follows:

- **Natural seated posture recognition:** We target subtle, natural seated postures observed during desk work rather than extreme postures.
- **SMPL-based trunk-aware features:** We demonstrate that SMPL-based 3D skeletal keypoints extracted from monocular RGB images outperform conventional 2D and sparse 3D skeletal representations (YOLO (Ultralytics inc.,), VideoPose3D (Pavllo et al., 2019)) for seated posture classification.
- **Integration with a smart chair:** We demonstrate the feasibility of integrating monocular posture recognition with physical posture guidance using a seat-angle adjustable smart chair.
- **Large-scale training and evaluation:** We evaluate the proposed method on data from 30 participants and conduct a preliminary user study with 10 additional participants.

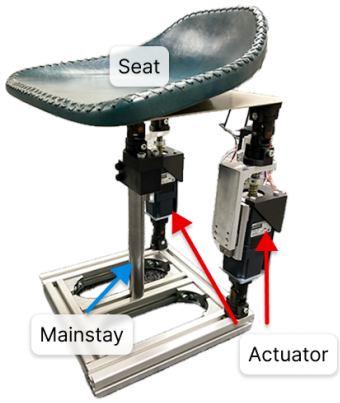


Figure 1: Appearance of the seat-angle-control smart chair (Esumi et al., 2024). The seat surface is tiltable in anterior–posterior and left–right directions (within $\pm 7^{\circ}$).

2 DATASET COLLECTION EXPERIMENT

The data used in this study consist of images and posture tracking data collected during desk work by Shimomae and Esumi et al. (Shimomae et al., 2025; Esumi et al., 2024), and no new data collection was conducted. In this chapter, we briefly describe the experimental procedure for data collection and the construction of the dataset in the previous studies.

2.1 Experimental Setup

Participants performed desk work tasks using a personal computer under the following two conditions: (1) *Seat-fixed condition*: the seat surface was fixed horizontally. (2) *Random seat-angle variation condition*: the seat surface was randomly changed among six representative tilt patterns (*forward4*, *forward2*, *back4*, *back2*, *left3*, and *right3*). The data from condition (1) were used for training seated posture classification (Chapter 3), and the data from condition (2) were used for training posture prediction for each seat tilt pattern (Chapter 4). Although the seat angle could be controlled continuously, six representative tilt patterns were adopted to ensure experimental consistency and safety during long-term use. Thirty university students (male and female) participated as subjects, and each condition was conducted for approximately 45 minutes (three 15-minute sessions), resulting in a total of 45 hours of data.

2.2 Sensor

Posture data were acquired using a combination of a motion capture system and webcams. The tracking



Figure 2: Three examples of seated posture. From left to right: "Good," "Slightly Poor," and "Poor.".

data obtained from the motion capture system were collected for the purpose of labeling the images. The motion capture system consisted of four infrared cameras configured as an optical tracking system, which captured the 3D coordinates of reflective markers attached to the participants at 100 fps. To accurately capture features essential for posture analysis, such as spinal curvature and pelvic tilt, 11 reflective markers were attached. Three webcams were mounted on the side of the participant to record the overall posture appearance at 30 fps.

2.3 Data Annotation

Annotations were performed on the data from the seat-fixed condition (1). For each participant, 100 still images were randomly extracted, and approximately 3,000 images in total were manually labeled by five annotators into three classes: Good, Slightly Poor, and Poor seated posture. Using the manually labeled data, an SVM classifier that takes motion capture tracking data as input was trained, and automatic labeling was performed on the remaining unlabeled data. The macro F1-score of the SVM was 0.94, indicating high classification performance. Examples of annotations are shown in Fig. 2.

3 SEATED POSTURE CLASSIFICATION

We propose a method that extracts SMPL-based 3D skeletal keypoints from monocular images and classifies seated posture into three classes using a GCN.

3.1 Proposed Method

In this study, we propose a method that extracts SMPL-based 3D skeletal keypoints from monocular RGB images using HMR2 (Goel et al., 2023) and classifies seated posture, such as in Fig. 2, into three classes using a Graph Convolutional Network (GCN). Previous studies have faced difficulties in adequately

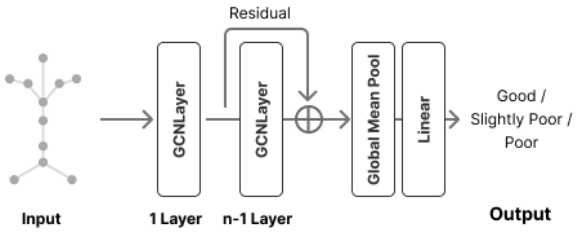


Figure 3: Overview of the classification method. 3D skeleton points based on SMPL are extracted from monocular RGB images using HMR2 (Goel et al., 2023) and classified into three seated posture classes using a Graph Convolutional Network (GCN).

capturing subtle differences in trunk alignment—such as pelvic tilt and spinal curvature—that are essential for seated posture estimation. To overcome these limitations, instead of directly extracting skeletal keypoints from images, we adopt an approach in which a 3D human body model is fitted to a monocular image, and dense skeletal keypoints are extracted from the estimated 3D body model.

3.1.1 3D Human Body Model

As the 3D human body model, we adopt SMPL (Loper et al., 2015), a statistical human body model that can uniformly represent body shape and pose. SMPL is characterized by its ability to generate natural human meshes from model parameters. Compared with conventional 2D/3D skeletal keypoint extraction methods (Kapoor et al., 2022; Li et al., 2023), the 3D skeletal keypoints obtained from SMPL can densely capture systematic 3D keypoints including the spine and pelvis, as shown in Fig. 4a. This allows detailed and quantitative representation of trunk alignment specific to seated posture. Figure 4a shows the joint configuration of SMPL. SMPL consists of 24 joints that represent the main anatomical structures of the human body.

3.1.2 Fitting of Human Body Models

As the method for fitting the SMPL (Loper et al., 2015) model to monocular RGB images, we adopt HMR2 (Goel et al., 2023). HMR2 (Goel et al., 2023) uses a Transformer as the backbone, which improves global context understanding of images and enhances robustness. It also introduces the Mesh Vertex Loss (PVE), enabling detailed reconstruction of body surface shape. This allows for accurate and stable estimation of SMPL parameters from monocular images. The ability to reconstruct detailed body surface shape is important for capturing subtle differences in trunk alignment in seated posture; therefore, we employ HMR2 in this study.

3.1.3 3D Skeletal Point Extraction

Using the publicly available pretrained HMR2 model, we estimate SMPL parameters from monocular RGB images and then extract SMPL joint 3D skeletal keypoints. In the transformation from image to SMPL, the information obtained from the image is normalized into the SMPL parameter space. In addition, when extracting 3D skeletal keypoints, we set all SMPL shape parameters to zero to obtain a standard body shape, thereby eliminating individual differences such as body size. Figure 4a shows an example of SMPL-based 3D skeletal keypoints overlaid on actual images. From this figure, we can see that spinal curvature and pelvic tilt are captured as 3D skeletal keypoints. Meanwhile, we observed that limb joints change regardless of posture, and therefore decided to exclude limb joint information.

3.1.4 Posture Classification

GCNs are suitable for learning graph structures such as 3D skeletal keypoints and can effectively capture dependencies between joints; hence, we use a GCN to classify seated posture. By stacking multiple graph convolution layers with residual connections, we aim to learn both local and global dependencies between joints and capture subtle differences in seated posture. Furthermore, we apply Batch Normalization and Dropout to suppress overfitting. For input data normalization, we translate the skeleton so that the pelvis is centered at the origin and scale it so that the distance between the pelvis and head becomes one.

3.2 Performance Evaluation Experiment

In this section, we describe the experimental procedure for evaluating the seated posture classification performance of the proposed method.

3.2.1 Evaluation Method

We used six-fold cross-validation for evaluation and performed training, validation, and testing on a per-participant basis in each fold. Only data that were manually labeled were used as test data. Participants were grouped into six groups of five subjects each. In each fold, four groups (20 subjects) were used as training data, one group (five subjects) as validation data, and one group (five subjects) as test data. As the evaluation metric, we computed the macro F1-score for each class.

3.2.2 Experimental Setup

The batch size was fixed at 1024, and the number of epochs was fixed at 1000 with early stopping based on the F1-score. The number of GCN layers was set to 6, the number of hidden channels to 512, and the dropout rate to 0.4. We used Adam as the optimizer and tuned the learning rate (lr) and weight decay using Bayesian optimization. The learning rate was set to 0.00004, and the weight decay was set to 0.001689.

3.2.3 Comparative Approach

To demonstrate the effectiveness of the proposed SMPL keypoint configuration focusing on 12 trunk-centric joints (NoLimbs), which excludes limb joints (hand, wrist, elbow, knee, ankle, foot), we compare the following three SMPL keypoint configurations:

- SMPL All: all 24 joints shown in Fig. 4a are used.
- SMPL NoHead: 10 joints obtained by excluding the head and neck from NoLimbs.
- SMPL NoSpine: 8 joints obtained by excluding Spine1 and Spine2 from NoLimbs, connecting the edge between Pelvis and Spine3.

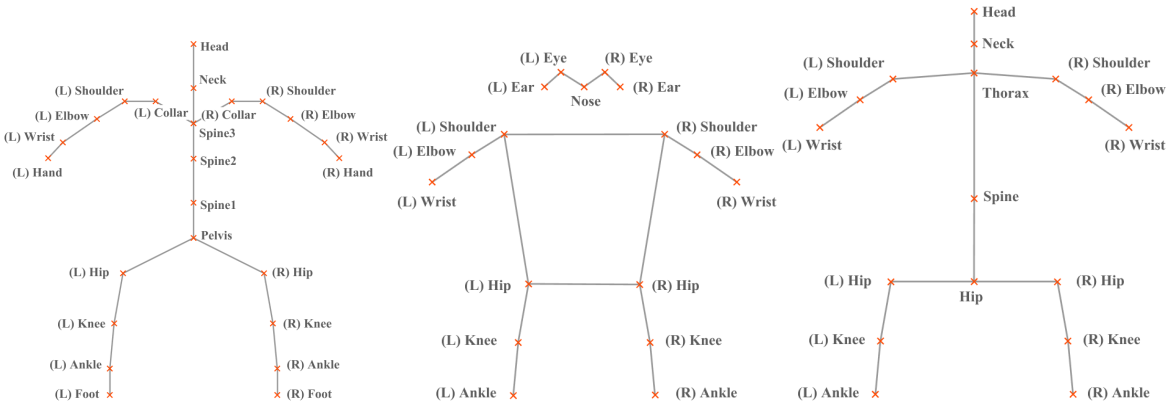
We also evaluate classification performance using other types of features, namely silhouette images, 2D skeletal keypoints, and 3D skeletal keypoints extracted by a method other than the proposed approach. Silhouette images and 2D skeletal keypoints from Yolo (Ultralytics inc.,) are extracted using the same procedures as those in Shimomae et al. (Shimomae et al., 2025). Silhouette images are intended to capture pure posture shape information by removing the influence of background and other factors. 2D skeletal keypoints constitute common features used in many conventional methods, and are adopted here as a baseline for comparison. 2D skeletal keypoints are extracted using Yolo (Ultralytics inc.,), a widely used method.

For 3D skeletal keypoints, we use VideoPose3D (Pavllo et al., 2019), which estimates 3D skeletal keypoints by considering temporal information from 2D keypoints extracted from images. The joint configuration is obtained by removing limb joints from the VideoPose3D skeleton in the same manner as SMPL NoLimbs, resulting in 9 trunk-centered joints.

3.2.4 Results

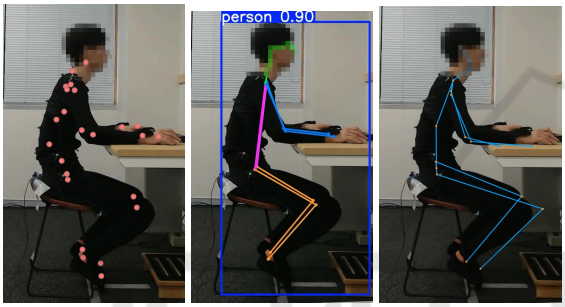
The seated posture classification results for each type of posture feature are shown in Table 1.

From the results, the SMPL NoLimbs joint configuration achieved the highest F1-score (0.595). Below, we discuss possible reasons for these results.



(a) SMPL. (b) Yolo. (c) VideoPose3D.

Figure 4: (a) Example of a 3D skeleton point structure based on SMPL (Loper et al., 2015). (b) Example of a 2D skeleton point structure based on Yolo (Ultralytics inc.,). (c) Example of a 3D skeleton point structure based on VideoPose3D (Pavllo et al., 2019). (R) stands for Right, (L) stands for Left.



(a) SMPL. (b) Yolo. (c) VideoPose3D.

Figure 5: Examples of skeleton points overlaid on RGB images. (a) SMPL-based 3D skeleton points projected onto the image plane. (b) Yolo-based 2D skeleton points projected onto the image plane. (c) VideoPose3D-based 3D skeleton points projected onto the image plane.

Table 1: Results of 3-class classification of seated posture.

Input Feature	Method	Ave F1
Silhouette image	ResNet50	0.589
YOLO	GCN	0.459
VideoPose3D (NoLimbs)	GCN	0.497
SMPL All	GCN	0.457
SMPL NoLimbs	GCN	**0.595**
SMPL NoHead	GCN	0.491
SMPL NoSpine	GCN	0.549

SMPL NoLimbs showed higher classification performance than YOLO and VideoPose3D. As shown in Fig. 5, SMPL-based features contain richer spinal and pelvic information than YOLO and VideoPose3D and thus more easily capture subtle differences in seated posture, such as spinal curvature and pelvic tilt. Furthermore, individual body shape differences were normalized into the SMPL parameter space by passing through SMPL and setting the shape parameters to a standard body type, which likely reduced the influence of individual differences that are unnecessary for posture classification.

The next best performance after the proposed method was obtained with silhouette images, with a classification performance of 0.589. Silhouette images directly reflect the posture shape in the image and are useful for side-view images as in this study; however, the lack of depth information and susceptibility to clothing effects may be reasons why they underperform SMPL-based features. In this study, all participants wore the same tight clothing, minimizing the influence of silhouettes; however, in real-world usage, the influence of clothing differences may become substantial.

Differences in classification performance due to joint configuration also became clear. SMPL NoLimbs uses trunk-centered joints including the spine, pelvis, and shoulder girdle, and was shown to effectively capture features of seated posture. On the other hand, SMPL All, which includes limb joints, exhibited lower accuracy, suggesting that limb joints acted as noise. Because limb states are not necessarily strongly related to seated posture, they may provide features that do not contribute to classification and hinder learning. The accuracy decrease of SMPL NoHead, which excludes the head and neck, suggests that head position and tilt also contribute to posture classification. Differences in seated posture are reflected not only in the angles of the pelvis and spine but also in head position and tilt. Similarly, the accuracy decrease in SMPL NoSpine, which excludes part of the spine, indicates the importance of detailed representation of spinal curvature.

These results show that for seated posture classification, the most effective approach is to use 3D information with a minimal yet stable set of trunk-centered

keypoints. In particular, utilizing stable 3D estimation methods based on body shape, such as SMPL, makes it easier to capture subtle differences in posture and improves classification performance.

4 SMART CHAIR INTEGRATION AND EVALUATION

In this chapter, we evaluate the feasibility of recognition-driven physical intervention by integrating the proposed seated posture classification method with a seat-angle adjustable smart chair (Esumi et al., 2024).

4.1 System Overview

As illustrated in Fig. 6, the system consists of three main modules: feature extraction, posture classification, and control intervention. First, SMPL-based 3D skeletal keypoints are extracted from monocular RGB images and normalized. The posture is then classified into three classes (Good, Slightly Poor, Poor) using a GCN. When non-Good posture is detected for consecutive frames, the system triggers a control intervention.

The design of the control framework aims to balance responsiveness and user comfort. Immediate seat motion in response to transient posture changes may cause discomfort or unnecessary intervention; therefore, control is activated only when poor or slightly poor posture persists for multiple frames. This design allows the system to respond to sustained posture degradation while avoiding excessive or unstable seat motion.

The control intervention adjusts the seat angle of the smart chair in anterior–posterior or left–right directions. By modulating the seat angle, the chair passively influences pelvic tilt and trunk alignment, inducing gradual posture changes without explicit instructions to the user. After a short holding period, the seat is gradually returned to the horizontal position to maintain comfort and safety. This closed-loop process enables continuous posture monitoring and adaptive physical guidance during desk work.

4.2 Control Strategy Based on Posture Prediction

To select an appropriate control command, we employ a posture prediction strategy that estimates the posture outcome corresponding to each candidate seat tilt. For a given posture state, the system predicts the expected posture after applying each control command and evaluates the predicted posture using the same posture classifier. The control command that is expected to yield the best posture class is then selected.

This prediction-based strategy avoids handcrafted rules that directly map postures to control actions. Instead, it provides a flexible framework that can adapt to different users and posture tendencies, while maintaining consistency between posture recognition and control evaluation. Although the current implementation uses predefined tilt commands, the framework can be extended to more adaptive control schemes.

4.3 Evaluation Experiment

4.3.1 Experimental Setup

A preliminary user study was conducted with 10 participants (aged 18–25) who did not participate in the dataset collection. Participants performed desk work under two conditions: a seat-fixed condition and the proposed adaptive control condition. Each condition lasted approximately 30 minutes, and the order was randomized. The sensing setup was identical to that used in the dataset collection.

4.3.2 Results and Discussion

Questionnaire results indicated that 80% of participants felt that their posture improved under the proposed system, while no participant reported frequent discomfort caused by seat motion. In addition, subjective reports of pain or fatigue in the neck, lower back, and hips decreased compared to the seat-fixed condition.

From observational analysis, the proposed system was effective for participants whose posture degradation was primarily related to pelvic tilt or trunk inclination. In contrast, posture patterns dominated by arm support on the desk or atypical lower-limb positioning were less responsive to seat-angle intervention. These results highlight both the effectiveness and the limitations of seat-angle-based physical intervention and suggest that combining posture recognition with additional contextual cues may further improve guidance performance.

Overall, this preliminary evaluation demonstrates the feasibility of recognition-driven physical intervention using a seat-angle adjustable smart chair and supports the practical applicability of the proposed approach.

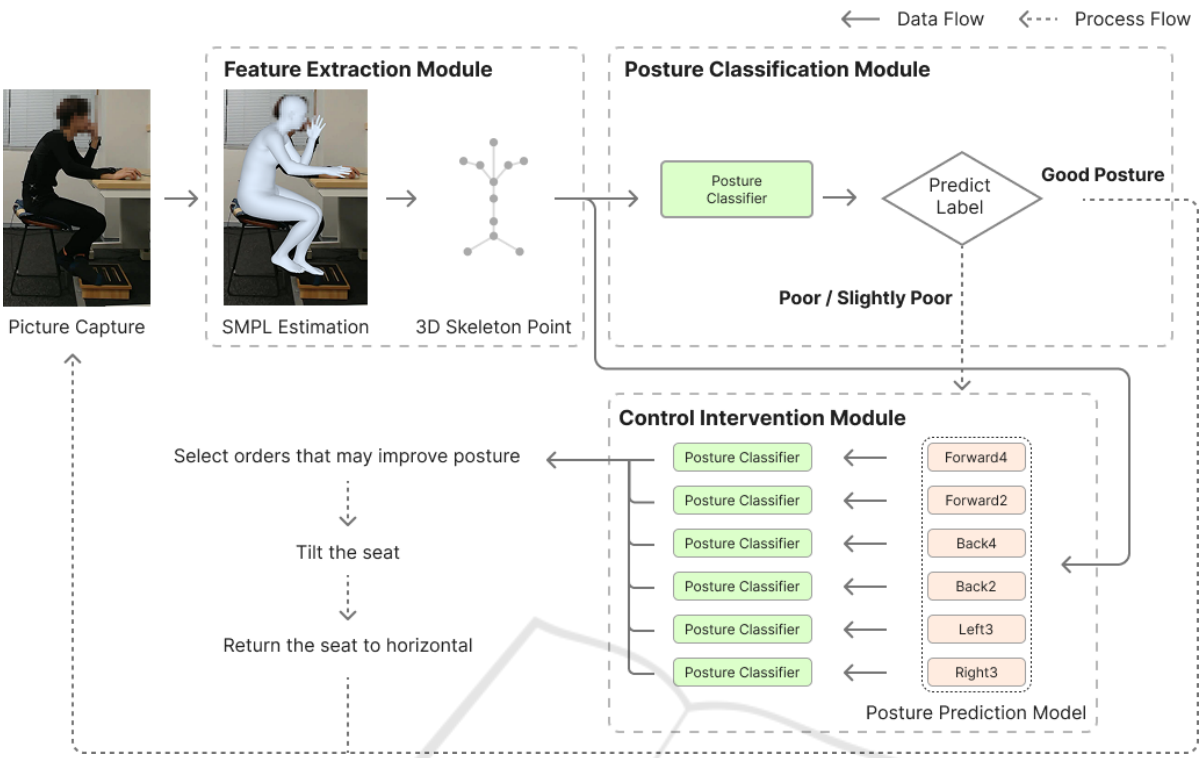


Figure 6: Overview of the integrated posture guidance system. SMPL-based 3D skeletal keypoints are extracted from monocular RGB images and classified using a GCN. When the posture is classified as Slightly Poor or Poor, an appropriate seat tilt command is selected and applied.

5 CONCLUSION

This study presented a monocular image-based seated posture recognition method using SMPL-based 3D skeletal keypoints and a graph convolutional network. By focusing on trunk-centered joints, the proposed approach achieved higher classification performance than conventional 2D and sparse 3D skeletal representations, demonstrating its effectiveness in capturing subtle differences in natural seated posture. Furthermore, we integrated the recognition method with a seat-angle adjustable smart chair and conducted a preliminary user study, which suggested the feasibility of recognition-driven physical posture guidance. Future work includes improving robustness under visual occlusions, personalizing posture guidance strategies, and evaluating long-term effects in daily-use environments.

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